2.00

LIVE

Damped Oscillation

$-kx$

⇒ The oscillations in which there is an extra force called as damping force that eats away the energy of the system

⇒ In one special case when damping force is proportional to velocity

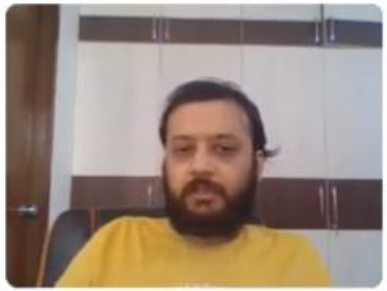
$$F_d = -bV$$

b = damping constant

unit of b is kg/sec

$$MLT^{-2} = [b]LT^{-1}$$

$$[b] = MT^{-1}$$



SA Sanskar Saxena
PICHLE SAAL BAHUT AAYE THE

RI Rishi Dalal
ok

SA Sanskar Saxena
SURFACE

AB Abhinav P.t
ok

SA Sanskar Saxena
M SE DIVIDE|

AB Abhinav P.t
yes

Type a message

Chat Doubts More

2.00

LIVE

$$F_{net} = -kx - bv$$

$$ma = -kx - bv$$

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0 \longrightarrow \text{II order DE}$$

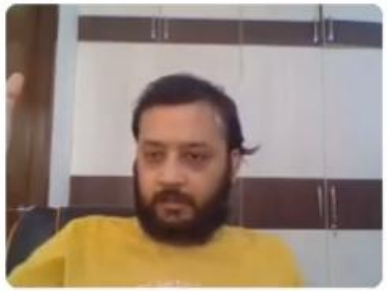
Depending upon the relation between m, b, k we have three possible cases

→ Underdamped $\longrightarrow b/2m < \omega_0$

→ Overdamped $\quad b/2m > \omega_0$

→ critically damped.. $\quad \frac{b}{2m} = \omega_0$

$$\omega_0 = \sqrt{k/m}$$



Rishav Raj
that may be a possibility😀

Abhinav P.t
😀

Rishi Dalal
sir so in our syllabus Fdamp is prop to velocity only?

Srijahnavi Ch
😀

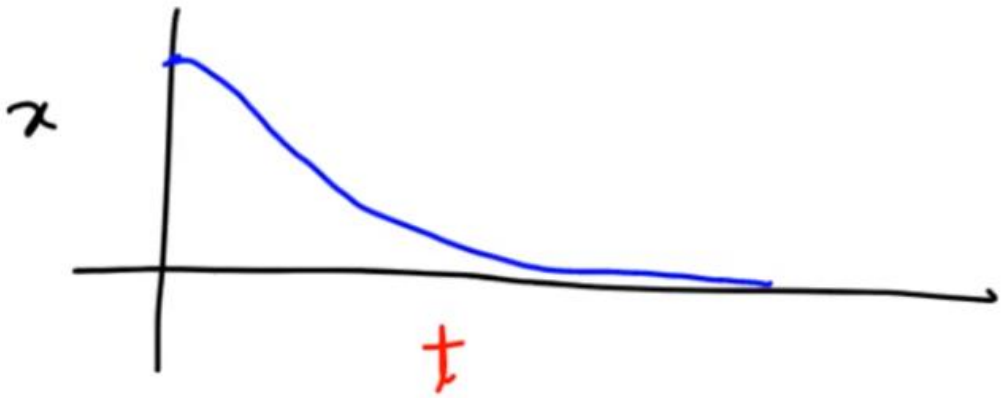
Srijahnavi Ch
yes

Type a message

Chat Doubts More

LIVE

Overdamped free



Body reaches its eqm position without oscillating slowly



Rishi Dalal
sir so in our syllabus Fdamp is prop to velocity only?

Srijahnavi Ch 😂

SR Srijahnavi Ch
yes

VA Vatsal Saxena
oh so like doors in malls and all?

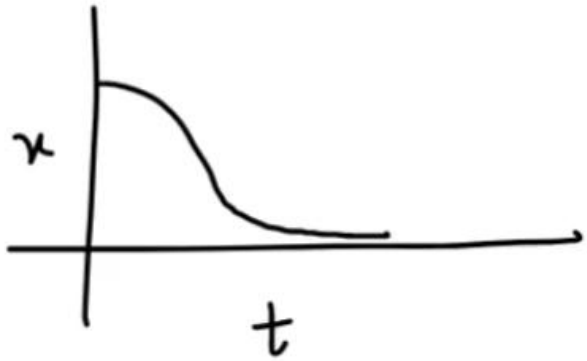
SA Sanskar Saxena
BHOOTIA BANGLE KA DARWAZA

Type a message

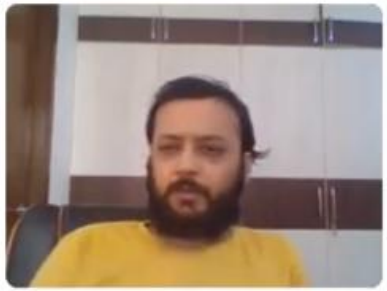
Chat Doubts More

LIVE

Critically Damped



In this case eqm is reached much faster in comparison to overdamped case.



Srijahnavi Ch: yes

Vatsal Saxena: oh so like doors in malls and all?

Sanskar Saxena: BHOOTIA BANGLE KA DARWAZA

Vatsal Saxena: so like squeezed of the previous case?

Sanskar Saxena: YES

Type a message

Chat Doubts More

LIVE

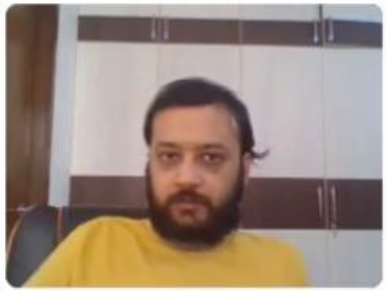
Underdamped Case

→ $\frac{b}{2m} < \sqrt{\frac{k}{m}}$

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

$$x = A_0 e^{-\frac{b}{2m}t} \sin(\omega t + \phi)$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$$



SR Srijahnavi Ch
yes

VA Vatsal Saxena
oh so like doors in malls and all?

SA Sanskar Saxena
BHOOTIA BANGLE KA DARWAZA

VA Vatsal Saxena
so like squeezed of the previous case?

SA Sanskar Saxena
YES

Type a message

Chat Doubts More

LIVE

Underdamped Case

$$\rightarrow \frac{b}{2m} < \sqrt{\frac{k}{m}}$$

$$m\frac{d^2x}{dt^2} + b\frac{dx}{dt} + kx = 0$$

$$x = A_0 e^{-\frac{b}{2m}t} \sin(\omega t + \phi) \rightarrow \text{learn}$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2} \rightarrow \text{learn}$$

ω = New Angular frequency

$T = \frac{2\pi}{\omega}$ = new time period

= 9ts constant



VA Vatsal Saxena
oh so like doors in malls and all?

SA Sanskar Saxena
BHOOTIA BANGLE KA DARWAZA

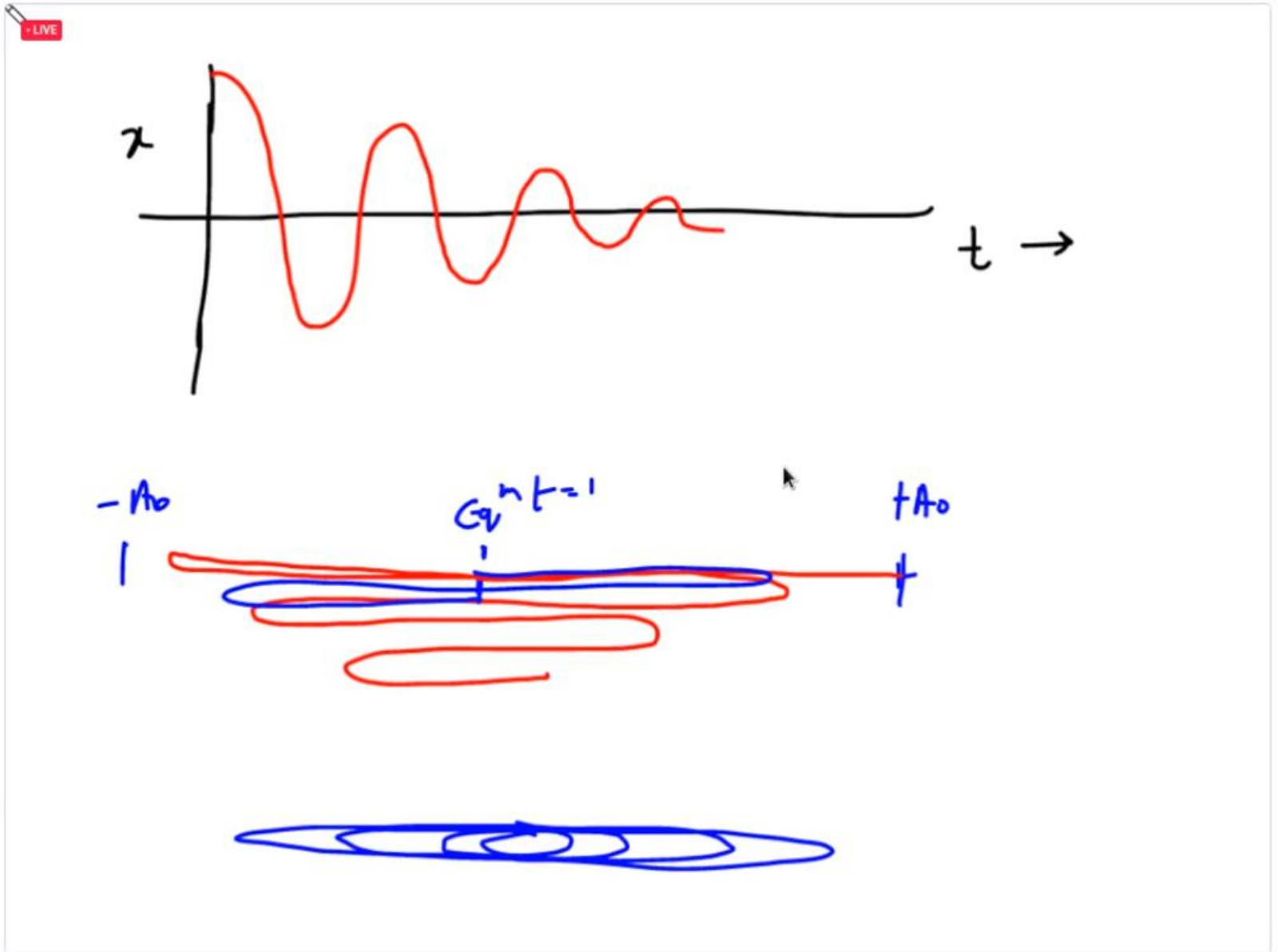
VA Vatsal Saxena
so like squeezed of the previous case?

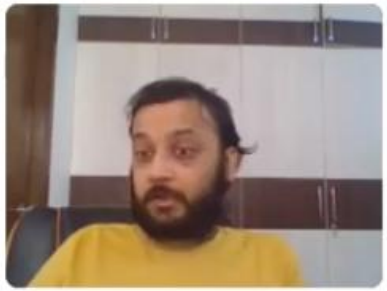
SA Sanskar Saxena
YES

AB Abhinav P.t
yes

Type a message

Chat Doubts More





AB Abhinav P.t
yes

SA Sanskar Saxena
amplitude decreasing

SA Sanskar Saxena
y

MO Mohit Singh
Yy

VA Vatsal Saxena
is the motion oscillatory sir?

AB Abhinav P.t
yes

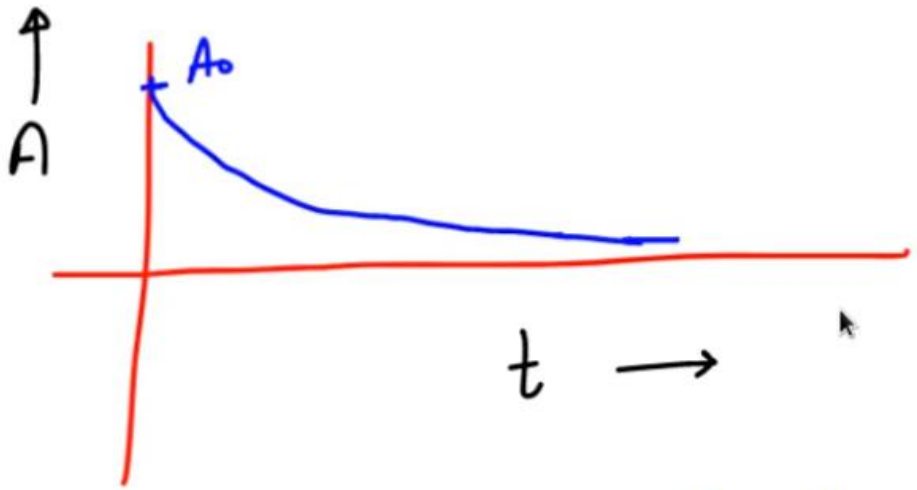
Type a message

Chat Doubts More

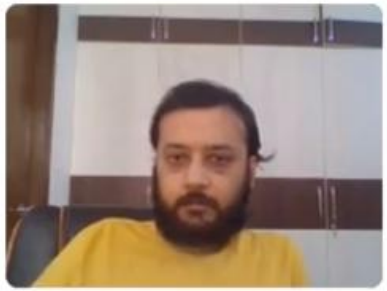
LIVE

$$A = A_0 e^{-\frac{b}{2m}t}$$

$$T_{1/2} = \frac{\ln 2}{b} \times 2m$$



⇒ Amplitude decreases with time



Yy

PR Pratham omkar Pat...
y

MA Manoj
Yes

SA Sanskar Saxena
yes

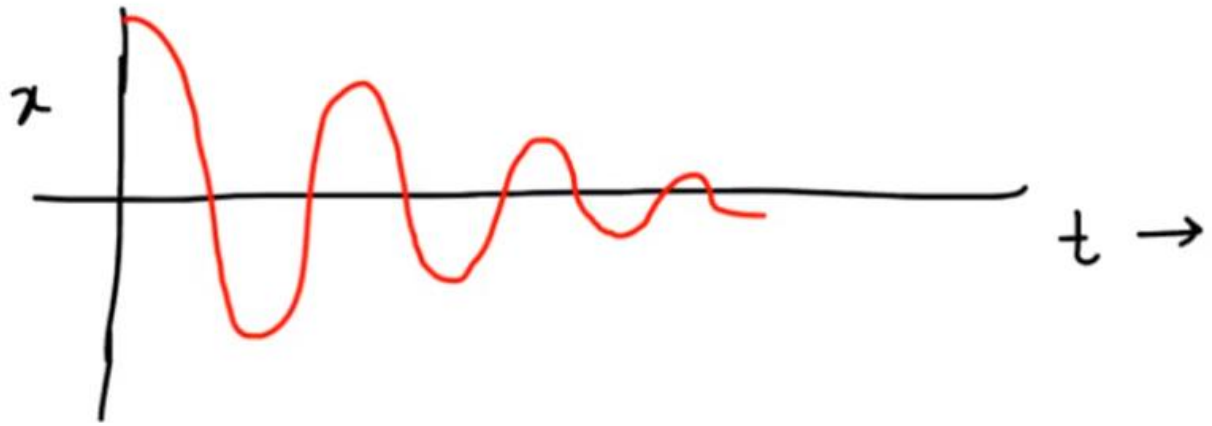
RI Rishi Dalal
no sir

AB Abhiharsh Chauhan
no

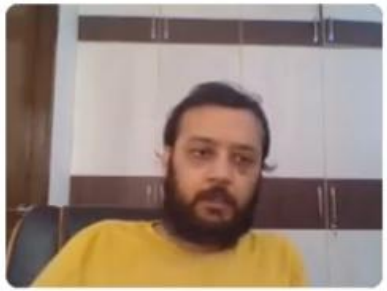
AB Abhinav P.t
n

Type a message

Chat Doubts More



⇒ Motion is not periodic, however time of oscillation remain constant



Yy

PR Pratham omkar Pat...
y

MA Manoj
Yes

SA Sanskar Saxena
yes

RI Rishi Dalal
no sir

AB Abhiharsh Chauhan
no

AB Abhinav P.t
n

Type a message

Chat Doubts More

LIVE

Energy in Damped oscillator

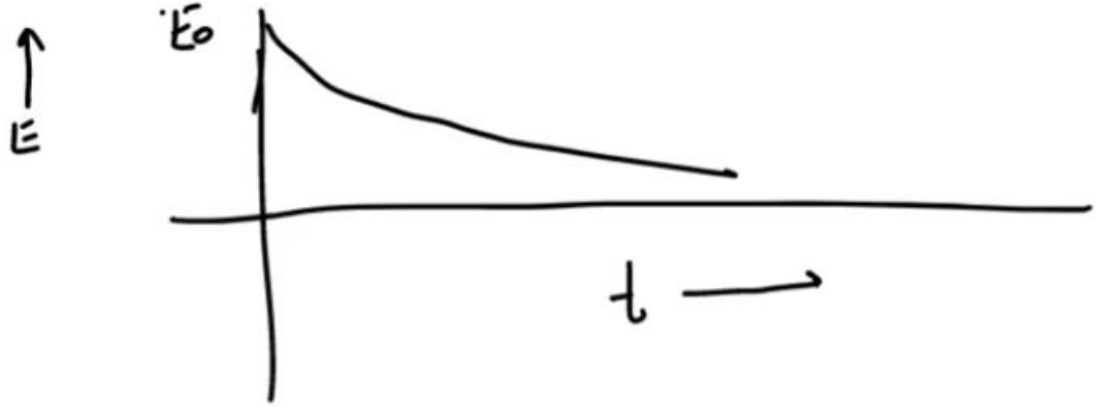
$$E = \frac{1}{2} k A^2$$

$$= \frac{1}{2} m \omega^2 A_0^2 e^{-\frac{bt}{m}}$$

$$E = E_0 e^{-b/m t}$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$$

$$T_{1/2} = \frac{\ln 2\, m}{b}$$





Samarth Nagpal
Omega not

AR Arnab Goyal
sir what is w here

AR Arnab Goyal
ok thank you sir

SA Samarth Nagpal
Ok

SA Sanskar Saxena
potential energy

Sanskar Saxena
□

Peeyush Jangid
□

□ □ 😂 😊 □ □

Type a message

Chat Doubts More

Rate of Loss of Energy or Power dissipated

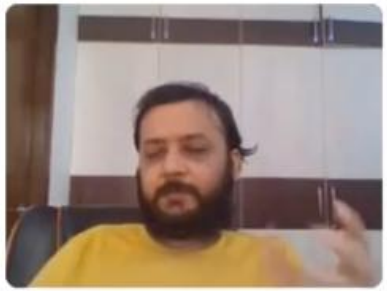
$$P = \frac{dE}{dt}$$

$$E = E_0 e^{-bt/m}$$

$$P = -\frac{E_0 b}{m} e$$

$$P = -\frac{b}{m} E$$

$$|P_{avg}| = \frac{b}{m} E_{avg}$$



Peeyush Jangid

AR Arnab Goyal
-dE/dt

AB Abhinav P.t
y

SA Sanskar Saxena
2 bhi to hoga

SA Sanskar Saxena
ok

SA Sanskar Saxena
got that

Type a message

Chat Doubts More

LIVE

Quality factor

Q is defined as 2π × ratio of average energy stored in one cycle to heat lost in that cycle.

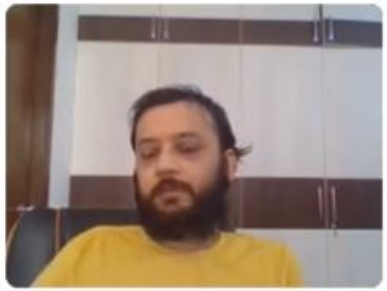
T = Time period

$$Q = 2\pi \times \frac{E_{avg}}{P_{avg} \times T} = \frac{2\pi m}{b \times \frac{2\pi}{\omega}}$$

$$= \frac{m\omega}{b}$$

$$\omega = \sqrt{\omega_0^2 - \left(\frac{b}{2m}\right)^2}$$

$$\simeq \frac{m\omega_0}{b} \quad \text{for small damping}$$



Chat

AR Arnab Goyal: -dE/dt

AB Abhinav P.t: y

SA Sanskar Saxena: 2 bhi to hoga

SA Sanskar Saxena: ok

SA Sanskar Saxena: got that

AR Arnab Goyal: yy

Type a message

Chat Doubts More

LIVE

A pendulum with time period of 1s is losing energy due to damping. At certain time its energy is 45 J. If after completing 15 oscillations, its energy has become 15 J, its damping constant (in s^{-1}) is: [2015]

$$E = E_0 e^{-\frac{b}{m}t}$$

$$15 = 45\, e^{-\frac{b}{m}\times 15}$$

$$\frac{1}{3} = e^{-\frac{b}{m}\times 15}$$

$$\ln\frac{1}{3} = -\frac{b}{m}\times 15$$

$$\boxed{\frac{b}{m} = \frac{\ln 3}{15}}$$



Pratik
ln3/15

AB Abhiharsh Chauhan
ln3/15

HA Hardik Sharma
ln/30

HA Hardik Sharma
ans ln3/30

RI Rishi Dalal
ln3/15

HA Harshal Sethiya
Ln3 /15

SA Sanskar Saxena
yes

Type a message

Chat Doubts More

LIVE

A block of mass 0.1 kg is connected to an elastic spring of spring constant 640 Nm^{-1} and oscillates in a damping medium of damping constant 10^{-2} kg s^{-1}, The system dissipates its energy gradually. The time taken for its mechanical energy of vibration to drop to half of its initial value, is closest to: |2017|

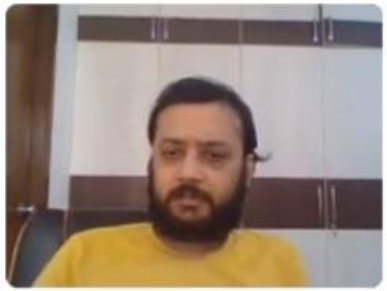
$b = 10^{-2}$

$E = E_0 e^{-\frac{b}{m}t}$

$T_{1/2} = \frac{\ln 2 \, m}{b}$

$= \frac{0.1}{10^{-2}} \ln 2$

$T_{1/2} = 10 \ln 2$



Rishi Dutta
7ec

AB Abhinav P.t
10ln2

AR Arnab Goyal
7

PE Peeyush Jangid
10ln2

HA Harshal Sethiya
7sec

PE Peeyush Jangid
6.93sec

SA Sanskar Saxena
10lm2

Type a message

Chat Doubts More

$\log 1000 = \frac{t}{2} \log 2$ $\quad t = \frac{6}{\log 2}$

A damped harmonic oscillator has a frequency of 5 oscillations per second. The amplitude drops to half its value for every 10 oscillations. The time it will take to drop to $\frac{1}{1000}$ of the original amplitude is close to:

$e^{\ln \alpha} = \alpha$

[2019]

2sec

$T_{1/2} = 2$

$2 = \frac{\ln 2}{b} 2m$

$A = A_0 e^{-\frac{b}{2m} t}$

$\frac{A_0}{1000} = A_0 e^{-b/2m\, t}$

$\frac{1}{1000} = e^{-\frac{\ln 2}{2} t}$

$\frac{1}{1000} = 2^{-t/2}$

Sanskar Saxena
6ln10/ln2

Peeyush Jangid

SR Srijahnavi Ch
yes

HA Hardik Sharma
approx 20sec

KA Kartik Kumar
6/log2

SA Samarth Nagpal
1/1024 times in 10 half lives, therefore approx 20 seconds

Type a message

Chat Doubts More

LIVE

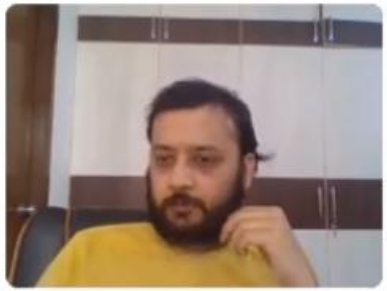
Q. The given system oscillates like a simple pendulum (Amplitude of oscillations is very small)

Find the time period of small oscillations.



m, r

$r \ll L$



Rishi Dutta
n

AR Arnab Goyal
no

AB Abhiharsh Chauhan
no

RI Rithvik Subash
nope

PR Pranesh K.r
no

KA Kartik Kumar
no

SW Swayam Prakash kar
no

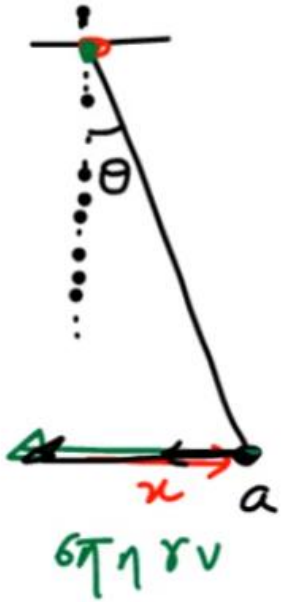
Type a message

Chat Doubts More

LIVE

$x = L\sin\theta$

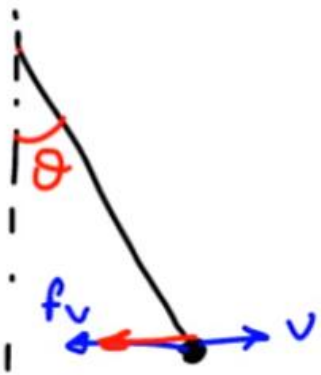
$-mgx \quad -6\pi\eta r v L = ML^2\frac{a}{L}$



$-mgL\sin\theta \quad -6\pi\eta r v L = ML^2\frac{a}{L}$

$6\pi\eta r v$

$ML\frac{d^2x}{dt^2} + 6\pi\eta r L\frac{dx}{dt} + mgx = 0$





JA Jashan Grover Insan
got it...aa gya

Samarth Nagpal

JA Jashan Grover Insan
force ki eq likh kr F=-kx

SW Swayam Prakash kar
log maje le rahen hai ab

SA Sanskar Saxena
yes

JA Jashan Grover Insan
yes...got it

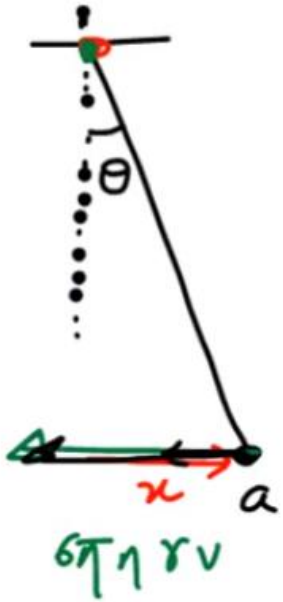
Type a message

Chat Doubts More

LIVE

$x = L\sin\theta$

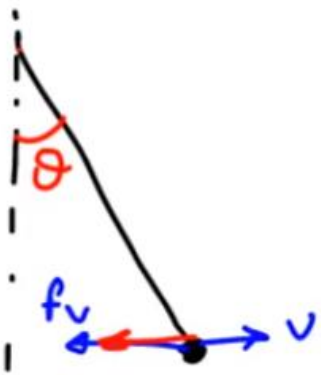
$-mgx \quad -6\pi\eta r v L = ML^2\frac{a}{L}$

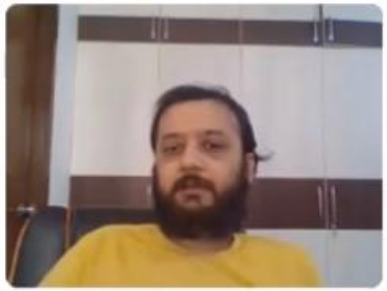


$6\pi\eta r v$

$-mgL\sin\theta \quad -6\pi\eta r v L = ML^2\frac{a}{L}$

$ML\frac{d^2x}{dt^2} + 6\pi\eta r L\frac{dx}{dt} + mgx = 0$





JA Jashan Grover Insan: got it...aa gya

Samarth Nagpal 😄

JA Jashan Grover Insan: force ki eq likh kr F=-kx

SW Swayam Prakash kar: log maje le rahen hai ab

SA Sanskar Saxena: yes

JA Jashan Grover Insan: yes...got it

Type a message

Chat Doubts More

LIVE

$$m\frac{d^2x}{dt^2} + 6\pi\eta r\frac{dx}{dt} + \frac{mg}{L}x = 0$$

$$\omega = \sqrt{\frac{g}{L} - \left(\frac{6\pi\eta r}{2m}\right)^2}$$

$$\omega = \sqrt{\frac{g}{L} - \frac{9\pi^2\eta^2 r^2}{m^2}}$$

$$T = \frac{2\pi}{\omega}$$

$$\omega_0 = \sqrt{\frac{k}{h}}$$

$$\omega_0^2 = \frac{k}{m}$$



Swayam Prakash kar
standard

Sanskar Saxena
b/2m ka form wala

Vatsal Saxena
rishi w rt(g/L)

Rishi Dalal
are ha root hota tha

Abhiharsh Chauhan
Sir aise 3-4 questions HW me dedo

Type a message

Chat Doubts More